MASSACHUSETTS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
2.671 Measurement and Instrumentation

Uncertainty Analysis Part 2: Curve Fitting

This second document was created to teach you how to analyze your data by fitting it to a mathematical model in a process known as curve fitting. Fitting experimental data to a mathematical model can be a useful technique to enable you to understand what the data mean. **This document describes how to select a curve fit model, how to interpret and display model parameter uncertainty, and a few techniques for how to measure "goodness of fit".**

Uncertainty from curve fitting reflects discrepancies between your data and your chosen model. The discrepancies may be due to measurement error, or the extent to which the model is an oversimplification of reality. In many cases, the curve fit model is based on the physics of the situation (e.g. Arrhenius form, exponential fit to first-order linear ODE system, complex exponential fit to second-order linear ODE system, etc.). In other cases, the model is a convenient mathematical function that represents the data well enough for tasks such as interpolation or comparison. In the latter cases, the mathematical function should be selected judiciously based on the expected application, and to avoid issues such as "over-fitting".

Curve fitting is a rich and sometimes complicated subject. This guide is merely an introduction to get you started with some common tools and techniques. Feel free to further your understanding of curve fitting with online research and discussions with peers and instructors.

This document can be found online in the Uncertainty page of the Wiki as a PDF file. This document assumes that you will use MATLAB, which will allow you to take advantage of the 2.671 MATLAB curve fitting building blocks, found here[*]. In addition, many curve fitting tasks can be rapidly accomplished in Logger Pro, which provides a good starting point when planning out your data analysis. We strongly recommend following the guidelines on the Wiki[†] to create useful MATLAB code for your data analysis, possibly after trying out various techniques in Logger Pro.

This text document includes detailed explanations and references. Essential information in this document has been extracted into a Slide Deck that should make it easier for you to understand and apply curve fitting techniques.

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[*] https://wikis.mit.edu/confluence/display/2DOT671/Curve+Fitting

[†] https://wikis.mit.edu/confluence/display/2DOT671/MATLAB

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1 Fitting to a Model

1.1 Steps to Select a Good Curve Fit

When selecting a model to fit your data, it is often necessary to take an iterative approach where you try several models and then evaluate and adjust each model before ultimately choosing the best fit for your data. It is recommended that you follow the process depicted in **Figure 1** and described in greater detail in this document.

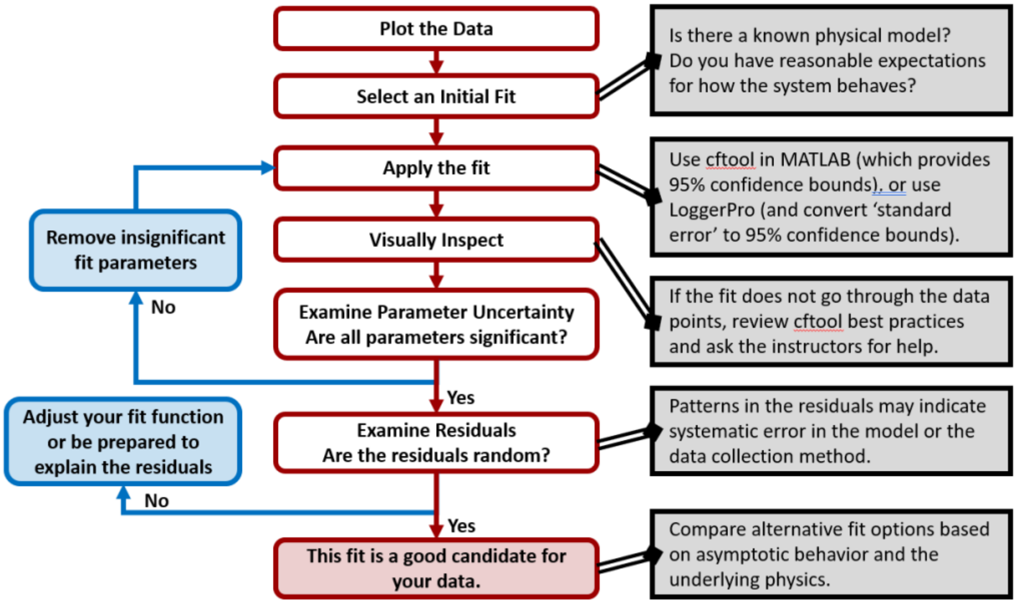


Figure 1: Flow chart describing the process for selecting, evaluating, and honing a curve fit model for experimental data. It is often necessary to look at several different model options before selecting the best candidate for your data.

1.2 Plot the Data: Anscombe's Quartet

Many experiments involve varying one parameter and measuring the effect on another. In many cases, the independent variable (y) exhibits a linear relationship to the dependent variable (x) and the measurement of interest is the slope (and/or the intercept) of a best fit line to the data.

Before fitting a line (or any function) to a set of (x,y) points, we strongly suggest **plotting** the data so that you can see if they look like they SHOULD be fit with a line (or selected function). Shown in **Figure 2** is "Anscombe's quartet": four datasets that have nearly identical statistics, but appear very different when graphed, only one of which is well described by the best-fit line [1].

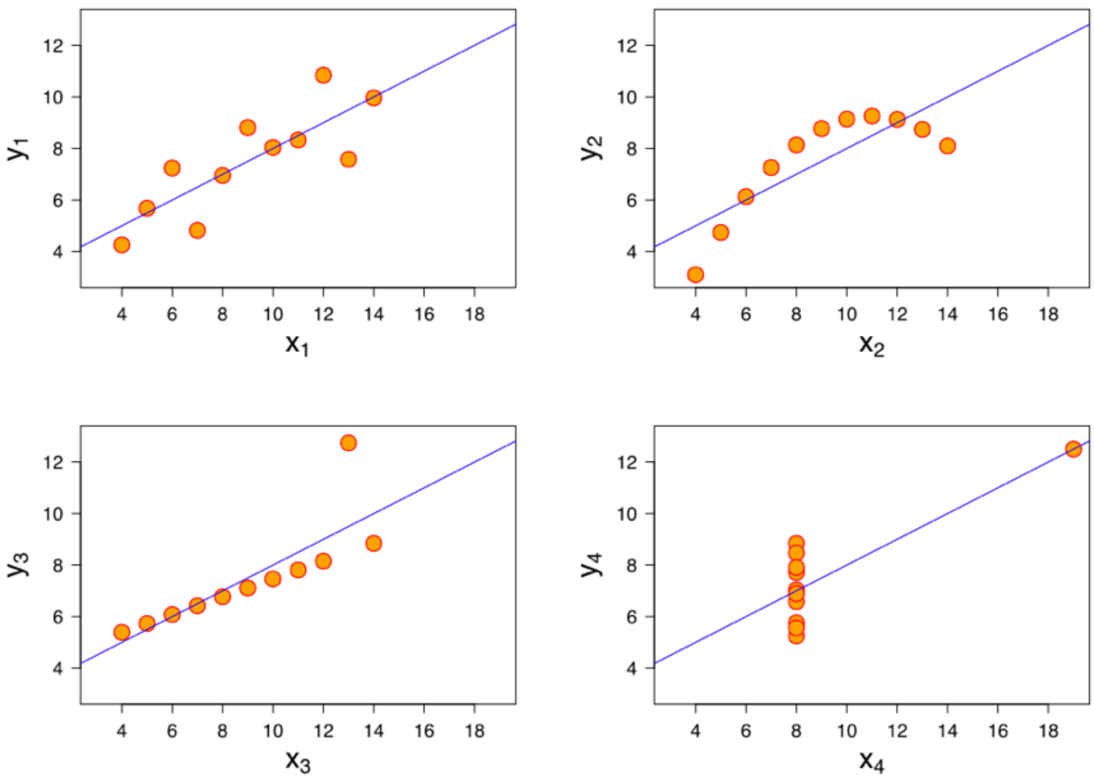


Figure 2: Graphs of the four data sets in Anscombe's quartet [2]. Statistical information for each data set is given in Ref. [1]. It is clear that only the top-left graph is well modeled by the linear fit line shown, even though all four have the same best-fit line with the same R^2, the same correlation between x and y, and the same mean and variance of both x and y. The top right graph appears to be a quadratic function, and the bottom left graph appears linear with a single outlier.

Most linear least-square algorithms minimize the vertical deviations (deviations in the y-directions) from the fitted line, and then the uncertainty in both the slope and the intercept depend on the ***standard error*** of the y-data about the fit [3]. There is in fact a closed-form solution for the uncertainty of the slope and intercept of a linear fit, but there is not for any other functional forms, so rather than showing you the closed-form solution only valid for a linear fit, we will teach you how to use the programs available to you to determine the best-fit parameters with their uncertainty.

You may be familiar with using "Trendlines" in Excel to fit data to a function. Unfortunately, parameter uncertainty for Trendline fits in Excel can only be obtained for linear fits. Do not use Excel for curve fitting; instead use MATLAB (preferably) or Logger Pro (less preferably) for all fits of data to any arbitrary fit function. Arbitrary fit functions are easily entered in both Logger Pro and MATLAB, and both environments also have a number of built-in functions.

1.3 Select a fit function

Statistician George Box is widely reported to have said "All models are wrong, but some are useful." As engineers and scientists, we often use mathematical models to represent the natural world. We combine measurements, theory, analysis, computation and judgement to describe a system or phenomenon, and use that understanding to achieve a useful purpose. For most interesting systems, high fidelity models that make accurate predictions are only achievable with complicated models. Fortunately, in many cases, a simplified model can provide "good enough" predictions to make useful progress without excessive labor building and validating the model.

For example, suppose you want a model to estimate trip duration for travel by bicycle. You might reasonably start with a proportional model assuming a constant speed, and divide the total trip distance by your average speed. You might reasonably add a constant or offset term to represent the walking time to get to your bike, unlock it, re-lock it at your destination. You might reasonably reflect that your cycling speed varies significantly with terrain, weather conditions (e.g. air temperature, wind speed and direction, etc.) and traffic levels. If you were using your model to estimate commuting time, terrain and traffic levels are known to you, and reasonably consistent. That might not be the case if the trip in question was a first-time trip in a new place. Now consider the task of using your model to estimate a long ride, such as the 300+ km Pan-Mass Challenge. For such an ambitious ride as this, fatigue and breaks for food and drink are likely to be a significant factor affecting your average speed. For a ride across the country, bicycle maintenance could also have a potentially significant effect on the duration of the trip.

The point of the foregoing example is to illustrate that the level of detail required depends on the purpose of the model. If you need to give your partner an ETA from Boston, a simple linear model is probably adequate. However, a simple linear model may not be adequate if you want to know the date for your return flight back from California following an Atlantic-to-Pacific ride.

The goal is to select the simplest possible model that adequately captures the physics of, and helps you explain and/or communicate the phenomena of interest in a useful quantitative way. When you set out to select a model, start by asking yourself whether there is an established physical model that you expect. Review your previous mechanical engineering classes, and talk to your instructors for inspiration. Examples include: Is there a chemical reaction that should take an exponential form according to the rate equation? Is there an object trajectory that should take a quadratic form in accordance with projectile motion? Is there a relevant thermodynamic or heat transfer correlation for the data?

Some real world processes are too complex for established physical models, but you can still make intelligent choices about what type of curve fit is appropriate. Ask yourself some questions like the following:

- Are there competing phenomena that could result in a local max or min in the data, or results in one or more inflection points in the data? This could indicate the need for a higher order polynomial. Beware the common pitfall of applying a 3rd or higher order polynomial fit to data that does not exhibit an inflection point – this will cause your model to have high error when extrapolated.

- Do I expect the dependent variable to approach a finite limit as the independent variable goes to infinity? This suggests a function with an asymptote, such as a logarithmic function or decaying exponential.

- Do I expect the dependent variable to continue to increase as the independent variable goes to infinity? This could suggest a linear, proportional, or exponential function.

1.4 Apply the Fit and Visually Inspect

MATLAB and Logger Pro both provides tools to quickly visualize and tweak curve fits. This section describes how to choose an initial fit and visually inspect the result, while Section 1.5 goes into greater detail on examining the curve fit parameter uncertainty with these same tools.

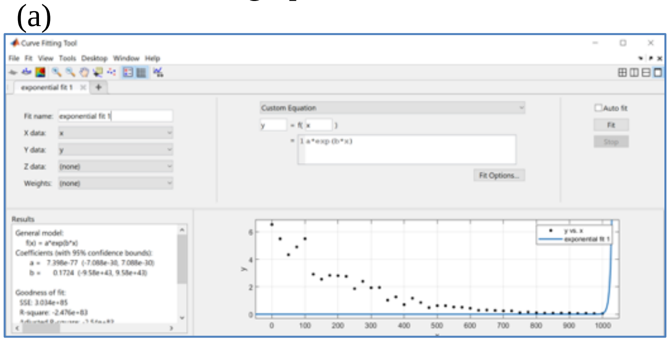
<u>MATLAB</u>: First, define your data in the MATLAB workspace. See 'Interpret → MATLAB → Reading Data Files' in the wiki if you are not familiar with how to do this. You can open the MATLAB curve fitting tool by typing the command 'cftool'. If you get an error message, make sure that the curve fitting toolbox has been installed.

In the Curve Fitting Tool window, give your Fit an appropriate name, and select the X data and the Y data from the dropdown menus. (If the variables you are expecting don't show up in the dropdown menus, it means they are not populated in your workspace). Confirm that the data populates the cftool graph as expected. This tool also allows you to select a region of interest, using the Tools – Exclude by Rule dialog box.

Now, select the desired equation from the dropdown menu. If you know the expected form of the equation, you may want to enter it as a 'Custom Equation' format. If 'Auto fit' is selected, you should see a blue line appear on your graph showing the resultant fit. (If 'Auto fit' is not selected, click the 'Fit' button to make the fit appear).

IMPORTANT: always visually inspect your fit to confirm that it makes sense and follows the data. Figure 3 depicts examples of a bad fit (a) and a good fit (b) using cftool. If cftool is not finding a reasonable fit for your data (especially common when using a Custom Equation), try some of the troubleshooting tips below.

(a)



(b)

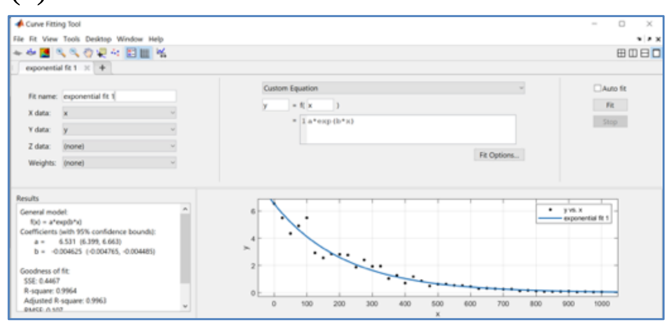


Figure 3: Curve fits generated using MATLAB cftool for data obtained from the equation $y = 10e^{-0.05x}$with added random error. (a) cftool failed to find a reasonable curve fit of the given exponential form despite the fact that such a fit exists. (b) providing Lower and Upper bounds for the Coefficients has allowed the curve fit tool to find an appropriate fit.

Troubleshooting tricks and tips:

- Under 'Fit Options', change the Algorithm to Levenberg-Marquardt.
- Under 'Fit Options', use the Trust-Region Algorithm, and provide known Lower and Upper bounds for the coefficients.

- If using a Custom equation, include scaling factors that cause the Coefficients (a,b,etc) to be of the same order, and the expected sign. (cftool solvers often struggle if the Coefficients are of wildly different orders).
- Talk to your instructors! Finding appropriate curve fits can be challenging, and we are here to help.

Logger Pro: See section 1.5.1 below for more information about curve fits in Logger Pro.

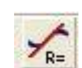
1.5 Examine Parameter Uncertainty

If you look at 'Case 3 Fit Parameters' under 'Uncertainty Analysis' on the 2.671 wiki, you will find a description of uncertainty analysis for Regression (Best-Fit) parameters. In cases where curve fitting is a productive data analysis technique, the parameter(s) of interest result from a functional relationship between two measured variables. For example, imagine sending a known current through a resistor and measuring the voltage. A plot of the measured voltage vs. the applied current should be a straight line. The slope of the best fit line gives the resistance. As another example, if a ball is thrown in the air, a plot of height vs. time should be a parabola. The coefficient of the 2nd order term can be related to the acceleration of gravity, g. We need a method to determine the statistical precision uncertainty of best-fit parameters. Once we have determined the precision uncertainty of best-fit parameters, we need to evaluate if the fit is statistically significant, i.e. if **all included parameters are statistically non-zero.**

1.5.1 Computation of Uncertainty for Linear Relationship

MATLAB and Logger Pro can both be used to compute the uncertainty of the slope and intercept of a fit of data to a line. The confidence interval is adjustable in MATLAB, and the default is 95% confidence, which is the 2.671 standard! Logger Pro lists the ***standard error*** of each parameter, which must be multiplied by the correct t-factor to obtain 95% confidence uncertainty, as described below.

We will start by describing how to use Logger Pro, and then MATLAB to obtain the slope and intercept with uncertainty for a linear fit.



Logger Pro: Highlight the region of interest in your data and press the Linear Fit button to perform a linear regression on the selected data. Double-click on the "helper object" showing linear fit parameters and correlation and check the "Show Uncertainty" check box.

Logger Pro automatically computes the "***standard error***" of each parameter, i.e. the 68% confidence uncertainty, analogous to $\{\sigma/\sqrt{n}\}$ for the uncertainty of the mean value. In order to obtain the 2.671 standard 95% confidence uncertainty, you must multiply the uncertainty values given by Logger Pro by $t_{0.025,\nu}$ which for small numbers of samples is found from the t-factor table defined in Uncertainty Analysis Part 1 and found on the Wiki (search for "t-factor table:"). The t-factor is approximately equal to 2 for $n > 30$. Remember that the degrees of freedom, $\nu = n - p$ where n is the number of points and p is the number of parameters (2 for a linear fit). For large numbers of points, simply multiply the listed uncertainty in both slope and intercept by a factor of 2 to obtain the 95% confidence interval. The **great advantage** of Logger Pro is that it is ***very easy to perform a function fit to a selected portion of the data***.

<u>MATLAB</u>: Use **cftool** in MATLAB, rather than basic fitting, in order to obtain the slope and intercept uncertainty. Like Logger Pro, cftool also allows you to select a region of interest, using the Tools – Exclude by Rule dialog box.

MATLAB by default lists the 95% confidence interval for each parameter in the format (m_{min} m_{max}) and (b_{min} b_{max}) for the slope m, and intercept, b, respectively. The corresponding 95% uncertainty in each parameter (u_m *and* u_b) is simply the difference between the upper and lower limit for each parameter divided by 2.

These calculations can also be done in MATLAB code using the fit() command, followed by coeffvalues() to unpack the curve fit parameters and confint() to compute the uncertainty. The MATLAB – Curve Fitting page on the Wiki contains several scripts that you can modify for your application.

1.5.2 Computation of Uncertainty for Higher-Order Relationship

Logger Pro and MATLAB also calculate uncertainty for higher-order fits. For Logger Pro, select the **Curve Fit** button  (immediately to the right of the **Linear Fit** button) and then select your desired fit function. You can also enter a custom fit function. In this case it automatically displays the uncertainty. <u>Remember that the displayed uncertainty for the fit parameters must be multiplied by the correct t-factor</u>, following the degrees of freedom guidelines given in Sect. 5.1.

In MATLAB using **cftool**, no adjustment of the listed uncertainties is required, and in fact you can adjust the width of your confidence interval as desired (i.e. obtain uncertainty for 95% confidence, 99% confidence, etc.). You can fit to arbitrary functions in cftool by scrolling up in the function list to "Custom Equation" and entering your desired function. You may need to use some of the cftool "tips & tricks" described in Section 1.4 above to make the curve fitting work for your data.

1.5.3 Goodness of Fit – Removing Insignificant Fit Parameters

There are numerous statistical methods to evaluate "goodness of fit" for the fit of a function to data, most commonly using the χ^2 (chi-squared) distribution [5]. In the interest of simplicity, we suggest starting with an alternative approach to determine the goodness of your fits in 2.671. Compare the uncertainty (with 95% confidence, as with everything in 2.671) of your fit parameters to the values of the fit parameters themselves. **If any one of your fit parameters has an uncertainty greater than its value**, that means that **zero** falls within the confidence interval for that parameter. In that case, it is not a statistically significant parameter, and should *usually* be removed from the fit function.

If you fit an established physical model, then one parameter being statistically indistinguishable from zero means there is an issue with the experimental measurements, or with the physical model. You might conclude that your measurements were not adequate, or that the physical model was not appropriate for your experiment. You could conclude that the physical model is wrong, when the physical model is sufficiently well-established, that is unlikely. For example, if you use video motion analysis to fit the position of a ball in free-fall at low speeds near the ground, your results should predict the acceleration due to gravity within 95% uncertainty. If they do not, a measurement error is more likely than an error in the value of g=9.81 m/s^2.

If you do not have an established physical model and you simply want a curve fit for interpolation or explanation and comparison reasons, you should not include terms that are statistically indistinguishable from zero. You should delete those terms, or select an alternative model equation that does not have statistically insignificant terms.

As an example, consider Fig. 4, the calibration curve relevant to the Soda Can. You will have measured the voltage output as a function of change in resistance. The relevant physical parameter for this fit is the slope, which is used to find the "calibration constant" needed to convert voltage output into resistance change and then, through the strain gauge calibration factor, the change in strain. As shown in the left graph of Fig 4, the data are extremely linear – in fact, the coefficient of determination, $R^2 = 1$, which you might assume means it is a "perfect" fit. However, the R^2 value has limited use in determining goodness of fit, as explained in Ref [4]. You can have a low R^2 value and still have a good model, or a high R^2 value for a model that does not fit the data [4]. The R^2 value for the proportional fit shown on the right in Fig. 4 is 0.9999 (less than for the linear fit), but as we explain below, this is a better model to fit to these data. For this reason, **we do not use R^2 in 2.671**; instead, we ask you to look at the uncertainty on the parameter values and use those to evaluate which model is a better fit to your data.

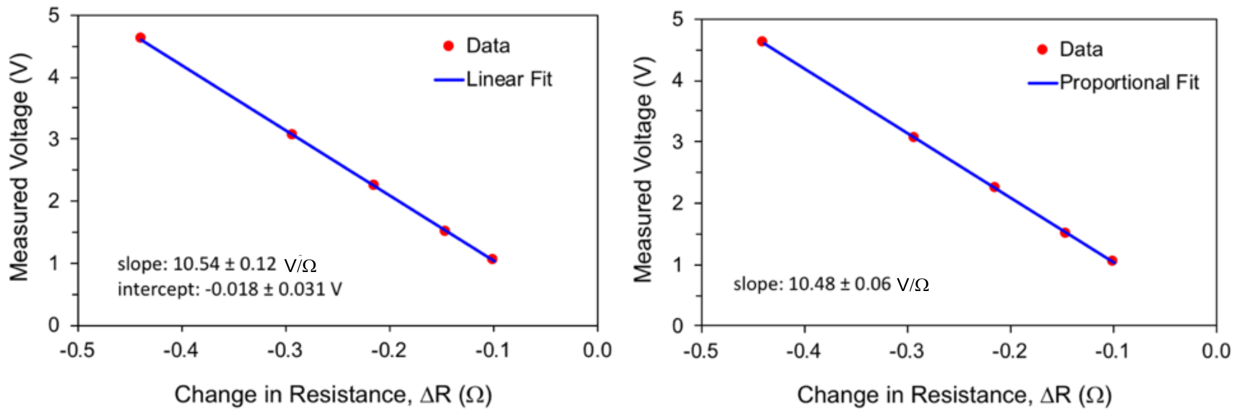


Figure 4: Linear (left) and proportional (right, intercept ≡ 0) fit to calibration data from the Soda Can experiment. The uncertainty on the fit intercept in the left graph is **greater than the value, indicating that the intercept is not statistically significant** (i.e., it could be zero, or positive, or negative). Fitting to a better model with no intercept term (a proportional fit) results in a factor-of-two reduction in the slope uncertainty!

The left graph of Fig. 4 shows the linear fit results in a fit where the **intercept uncertainty is greater than the intercept value**. This means that you cannot say with 95% confidence that the intercept is not zero (or in fact, whether it is positive or negative). Therefore, the constant term in the fit is not statistically significant, and should be removed. Re-fitting the same data to a proportional function (linear with no intercept) in fact reduces the slope uncertainty by a factor of two, indicating that you have higher confidence on the value of the slope (the relevant parameter for this example) with a proportional fit, and therefore it is a much better model to the data.

1.6 Model Error – Examining Residuals

Model error, also known as residual, is defined as the difference between the observed value and the value predicted by the model:

$$residual = model\ error \equiv observed\ value - predicted\ value$$

You can judge the goodness of fit of a model by comparing the total error of all data points fit by the model. However, some care is advisable in calculating the "total error". With a simple algebraic sum of the error terms, negative errors (over-predictions) and positive errors (under-predictions) could cancel each other out and result in a very small model error despite arbitrarily large errors in every point. The Least Absolute Residuals (LAR) overcomes this problem by calculating the total error as the sum of absolute value of error terms. A more common technique for aggregating the error is the Root Mean Square Error (RMSE). RMSE is calculated from the square root of the sum of the squares of individual error term. This converts positive and negative error terms to positive error, and due to the properties of the quadratic, it penalizes larger error terms more than small error terms.

An example of comparing residuals for two different models can be found below in Section 2 Linear vs. Quadratic Fit Example.

1.7 Additional 'Goodness of Fit' Measures and Techniques

1.7.1 Coefficient of Determination (R^2)

You may be familiar with the coefficient of determination, commonly known as the R^2 (R-squared) value. It is commonly reported by Excel, Logger Pro and MATLAB along with the curve fit parameters.

The coefficient of determination measures goodness of fit. In essence, it compares how well the model fits the data compared to how well a simple $y=b$ constant value model fits the data. Unfortunately, **R^2 is not a very good measure of goodness of fit for most engineering purposes**. An arbitrarily high-order polynomial model will often achieve a very high coefficient of determination for a sample of data. However, that same polynomial can then give wildly inaccurate predictions for new samples of data taken under the same conditions. In most engineering applications, you would expect that if you took new data, it would closely agree with the previous data and model. A high R^2 value only indicates that the model closely follows the data you have now. A high R^2 value does not indicate whether the model fits the situation, or if the model is simply "over-fit", where too many parameters or coefficients are used for the situation.

1.7.2 Adjusted R^2

To account for some of the shortcomings of conventional R^2, there is a quantity known as Adjusted R^2, which is a simple numerical adjustment of the R^2 value based on the number of data points, and number of parameters in the model. It attempts to measure whether the increase in R^2 from adding an additional parameter is greater than would be expected by chance. Adjusted R^2 is most useful for comparing related models, such as a quadratic ($y = A + Bx + Cx^2$) and linear ($y = A + Bx$) models. An alternative method for comparing fits with different numbers of parameters is to examine the overlap of the confidence intervals, as discussed in Section 3.6.

1.7.3 Model Cross-Validation

A more advanced technique for judging model goodness of fit is called cross-validation. With this technique, you randomly assign your observation data into two groups, the fit group and the validation group. It is common to randomly select 90% of your observation data in the fit group, and reserve the remaining 10% for cross-validation. Then you create a model fit for the fit group.

You use the resulting model to predict what you expect to see for the data in the validation group. By measuring the model error of those predictions, you can evaluate whether the model is likely to provide a good prediction of new observations, or whether it fits the observation data by chance.

1.7.4 Comparing multiple models

You can calculate the model error of various curve fit types or model equations to directly compare their performance or suitability.

There are many other means by which you might reasonably judge the goodness fit between your model and the data, or between two plausible model candidates The F-test for nested models is a statistical test that can indicate if one model provides a statistically significantly better fit than another related model.

1.8 Engineering Judgment

The extent to which you must measure or demonstrate the goodness of fit depends on the application. Some industries such as aerospace or medicine may have rigorously defined and enforced statistical norms and procedures. For your 2.671 Go Forth and Measure project, you are encouraged to apply engineering judgement, and think about how your model and its goodness of fit address the following issues:

1. The curve fit model is a valid physical model based on background research or generally accepted academic sources
2. The curve fit model has the correct asymptotic behavior for reasonable extrapolation
3. The parameter uncertainty is significantly less than the parameter value (See sec. 5)
4. The model residuals show no systematic variation*

* In some cases, you might use the presence of systematic variation in residuals to argue that the model is deficient and lacks some fundamental physics. For example, at high projectile velocity, a model that lacks aerodynamic drag would be expected to have significant, systematic residuals. A corollary to this is that for low projectile velocity, a model that includes aerodynamic drag would might display significant relative uncertainty of the model parameters concerned with aerodynamic drag because they do not exert a significant effect on the model.

2 Linear vs. Quadratic Fit Example

In the Lab 1 Postlab Part 1 we show you how to decide whether a proportional fit ($y = mx$) or a linear fit ($y = mx + b$) was a better model for your data, by examining the parameter uncertainty. This is a very straightforward example, because the uncertainty in the slope can be directly compared in the two cases. However, what if you are unsure if your data should fit a line or a quadratic? How do you determine which function is a better fit, and why?

We suggest the following process, previously depicted in Figure 1, to decide between linear and quadratic:

1. Plot the data.
2. Examine the plot and decide what fit function(s) you think would produce a good fit, using your physical intuition for the system as a guideline.

3. Try a fit function and visually inspect that the fit tool applied the function correctly.
4. Examine the parameter uncertainties. If any parameters are not statistically significantly different from zero (i.e. the uncertainty is larger than the value, so that the 95% confidence interval on this parameter includes zero), remove this parameter from the fit function (This is what you did in the Soda Can postlab when you went from a linear fit (with intercept) to a proportional fit (no intercept).
    a. Re-fit the data to modified fit functions until all parameters are statistically significant.
5. Examine the fit **residuals,** defined as the difference in y-values (**residual ≡ data – fit**) at each x value. Fit residuals should appear randomly distributed around zero. If residuals instead have a systematic relationship with x, as shown in **Figure 5 (d)**, the assumed fit function is not accurately representing the variation in the data with the independent ("x") parameter.

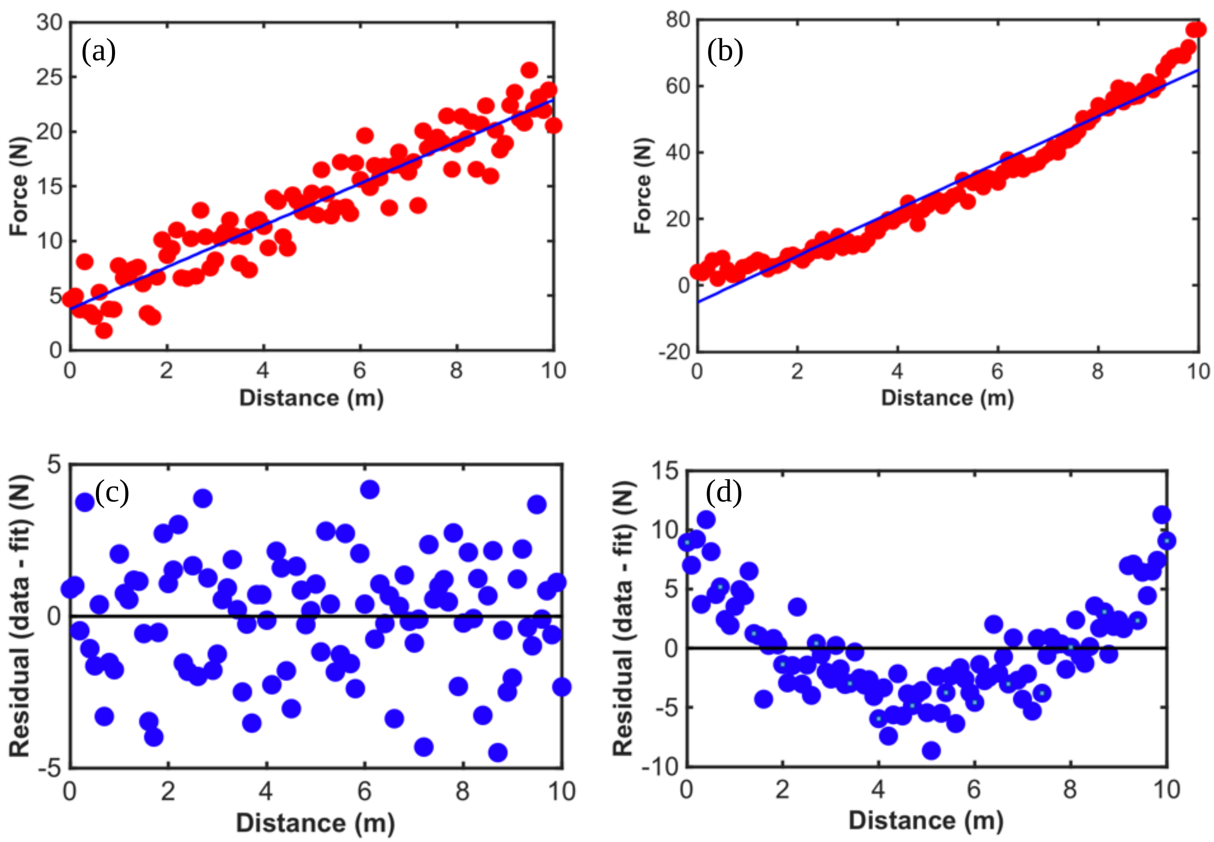


Figure 5: Linear fit of a noisy dataset that is (a) linear (b) quadratic. The fit residuals for the (c) linear dataset are randomly distributed about zero. However, the residuals for the (d) quadratic data have a clear systematic trend, indicating that the linear fit is not the correct fit function for this dataset. This is also apparent from the systematic variation of the data around the fit line in b).

6. If needed, adjust your fit function, ideally until the residuals show little to no evidence of a systematic dependence on the x-parameter, while ensuring that all parameters are statistically significant. If you cannot find a fit function that produces randomly distributed residuals, you will need to discuss the limitations of your fit model.

7. If you are left with a number of different fit functions that provide a good fit, you need to evaluate them based on their asymptotic behavior and your understanding of the underlying physics. Review Section 1.8 Engineering Judgment, and discuss with course instructors for additional guidance.

3 General fitting instructions

3.1 Fit all of your data, not just average points

In the interest of clarity, you may wish to simplify a figure and show average datapoints with error bars representing the confidence interval for that data, rather than showing all of the individual data points. However, when you fit a model to your data, **you should fit to all of the individual data points**, not just the average points. Including all of your data will yield a model that is a more complete representation of your data. Moreover, fitting to all of your data points not just the average points will almost always reduce parameter uncertainty of the model. An example is given in Figure 6.

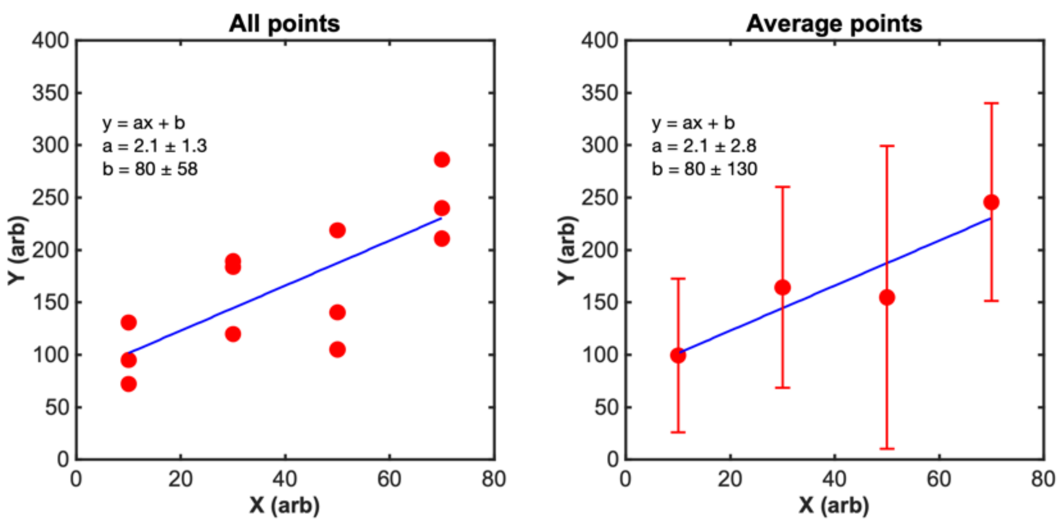


Figure 6: Example showing the statistical advantage of fitting to all your data points, rather than the average points. In the left graph, both slope and intercept parameters are statistically significant (non-zero). In the right graph (fitting to the average value of the 3 data points at each X value) neither fit parameter is statistically significant.

You must use your engineering judgment when deciding what graph to present in your technical communications: (1) show the fit with all the data points (left graph of Figure 6), (2) show the fit <u>generated from all the data points</u> on the same graph as the average points with error bars (graph similar to the right side of Figure 6 but <u>with the fit from the left side of Figure 6</u>).

For small number of points, choice 1 may be preferable, as the small t-factor results in large error bars. For larger number of points, consider choice 2. You must choose based on which option will be the most clear to the reader.

3.2 How to exclude outliers

There are many reasons why a measurement may be considered invalid and excluded from subsequent analysis. In some cases there was an unambiguous mistake in the intended experimental process, material or procedure (e.g. a sample broke, equipment malfunction, equipment not properly configured, etc.). Those data points can be annotated in your notebook with justification and elaboration, but excluded from analysis. In other cases, the root-cause of an unexpected measurement is not apparent to the researcher. It is a matter of engineering judgment whether the discrepant data are actually invalid outliers, or merely measurements whose cause/origin is not adequately understood.

It is not acceptable to omit outlier data entirely. It is preferable to include the data in preliminary analyses, with appropriate annotation and discussion, and omit it from later analysis. It is typical to see a curve fit to all data, and a refined curve fit with outliers excluded. In those cases, the criteria and rationale for rejection must be explicitly documented.

The foregoing does not mean that you have to present every and all measurements that you take. The point is that you must have, and explain, justifiable, repeatable and documented reasons for excluding data from a set. You should discuss the reasonableness of your exclusion criteria with 2.671 staff and your Lab Professor.

MATLAB's *cftool* has a convenient interface for interactively excluding outliers from a curve fit. This can help you build an intuition for the severity of outliers under different model and data conditions.

3.3 Partitioning a dataset

Your raw data may be a stream of measurements of which a small subset is valid or interesting. For example, you may collect flow measurements over a range of flow rates, and treat laminar flow differently than turbulent flow. In that case you would need to partition your dataset to apply the correct model in each case.

Another example would be discarding measurements that occur above or below an activation threshold (e.g. for a semiconductor), or at time points before or after the experimental measurement was made. Logger Pro allows you to do this graphically with Strike Through. cftool allows you to do this with the Tools – Exclude by Rule dialog box. The 2.671 Wiki has an example of doing this in MATLAB with a "Boolean Mask".

You may also consider **"piecewise" fits**, for example, a stress-strain curve may exhibit two linear regions with different slopes. You may perform linear fits separately on the two regions, and use the slopes to describe system behavior in the two regimes

In any case, your criteria for partitioning your data should be clearly documented and discussed as part of your method.

3.4 How to show parameter uncertainty

After you have completed the model fitting process, you should express the parameter including uncertainty and appropriate units. The 2.671 convention is 95% uncertainty, and you should analyze the fit equation to determine appropriate units for each model parameter or coefficient. Note that for mathematical functions such as $\ln(x)$, e^x, $\sin(x)$, the argument is unitless. Therefore, a model such as $y = A \ln(Bx + C) + D$, y, A and D would have the same units, C would be unitless and B would have the reciprocal units of x.

Reminder of how to report numerical results from Uncertainty Analysis Part 1, Sect. 5:

Every numerical result must include 3 components, as shown in Figure 7: value, uncertainty, and units.

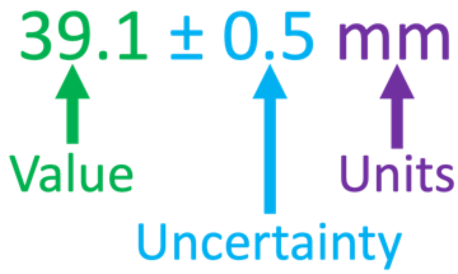


Figure 7: Required elements for reporting numerical results. The number of significant figures for the value and uncertainty are determined by the 2.671 Sig Fig Rules, listed below.

The main point of the sig fig rules is that you must **start with the uncertainty**, write it with **no more than two sig figs** (more would be meaningless information), and then **express the value of the parameter to the same numerical precision** (i.e. the same digit).

The sig fig rules only apply when you are reporting your final numbers; you should <u>never truncate numbers during a calculation</u> as that can cause round-off error, resulting in 10% or more error in the final result.

1. 2.671 Sig Fig Rules

1. Your computed uncertainty should be reported with **at most 2 significant figures**
2. Your value should be expressed to the **same numerical precision** as your uncertainty, i.e. to the same lowest digit

2. Examples:

3. $4.652 \pm 0.053\ m/s$
4. $(3.45 \pm 0.24) \times 10^{-6}\ V$
5. $(2.455 \pm 0.003) \times 10^{-5}\ m/m\ K$

6. ***Examples 2 & 3 (scientific notation):*** *value and uncertainty must be expressed to the same power of 10, and enclosed in parentheses as shown. Please format scientific notation as shown,* ***not*** *as E-06 or 10^(-6)*

7.

8. ***Example 3 (1 sig fig on uncertainty):*** *since the uncertainty is so much smaller than the value, only one digit of uncertainty is needed*

3.5 Calculations involving model parameters

Suppose that you fit a quadratic $y = Ax^2 + Bx + C$ model to some concave-down experimental data. Now you would like to calculate the x value corresponding to the maximum. You would expect the location of the maximum to be at $x = x_{max} = -B/2A$. You know from fitting the quadratic model to your experimental data that there is some uncertainty with parameters A, B, and C. What is the uncertainty of x_{max}, based on the uncertainty in parameters B, and A?

You may be tempted to apply the familiar rules for propagation of uncertainty by partial derivatives. Your instincts are correct but you must exercise caution. You may still use propagation of uncertainty, but you have to include the effect of covariance between model parameters. That is, when model equations are fit to observation data, the model fit parameters are not expected to be independent from each other. If they were independent, their covariance would be zero. That is, propagation of uncertainty must always include covariance. In cases where the terms are known to be, or may reasonably be approximated as, independent, the covariance is zero and the equations simplify to our familiar formulation of partial derivatives. Fortunately, MATLAB can calculate the full covariance matrix, and computations involving covariance are not prohibitively complicated. Figure 8 shows a calculation of the uncertainty of x_{max} using covariance as determined in MATLAB (covar_predint.m), which is built around the MATLAB function 'confint'. Another script on the 2.671 wiki (shade_confidence_interval.m) allows the user to plot the confidence interval associated with the model, as shown in the next section.

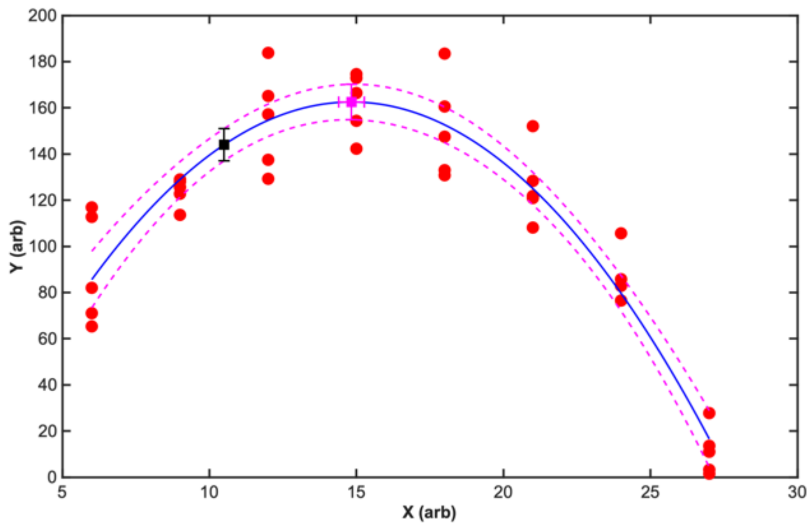


Figure 8: Plot produced using the covar_predint.m script from the 2.671 wiki on data generated from a quadratic function with added random noise. This script relies on the built in MATLAB function 'confint' to determine the confidence interval associated with a curve fit where the fit parameters are not independent variables, and uses computation of the Jacobian matrix from the fit process to determine the uncertainty on values derived from model parameters, such as the location and value of the maximum with uncertainty in both directions (magenta point with dual error bars) and the fit value at a given x-value (black point with vertical error bars) The 95% confidence bounds (magenta) around the curve fit (blue) can be understood as a function of the input x parameter.

3.6 Comparing models for two data sets

The shade_confidence_interval.m script is a useful tool for comparing multiple data sets across a range of input values. When plotted on the same graph, the models for two different data sets can be said to be statistically significantly different with 95% confidence if the confidence bounds (magenta dashed lines in Figure 8) do not overlap. With this approach, it is possible to identify regions in which two models converge or diverge, such as the example in **Figure 9** showing the bending strength of 3D printed beams with different printing patterns, as a function of infill percent.

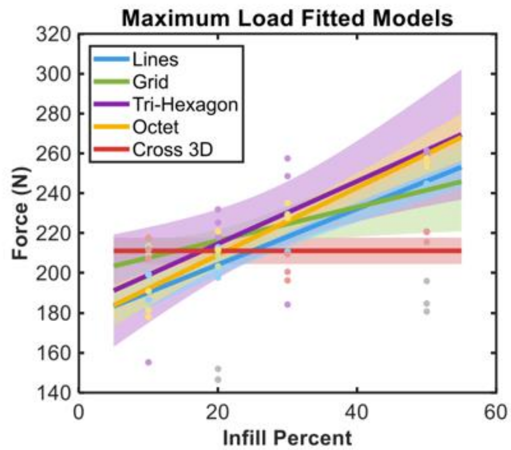


Figure 9: Effective use of the shade_confidence_interval.m script from "**Selecting Infill Pattern & Density for 3D Printing"**, Go Forth and Measure poster by 2.671 student Diane Heinle in the fall of 2022. The overlap of the shaded uncertainty regions around the model fit for each printed shape show that the models do not exhibit statistically significant different behaviors, except for the Cross 3D shape at high infill density**.**

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References

[1] Anscombe, F.J. "Graphs in Statistical Analysis", The American Statistician, Vol. 27, No. 1 (Feb., 1973), pp. 17-21

[2] From Wikimedia Commons, Attribution: by Anscombe.svg: Schutz, , accessed Jan 31, 2018. [https://commons.wikimedia.org/wiki/File:Anscombe%27s_quartet_3.svg - file](https://commons.wikimedia.org/wiki/File:Anscombe%27s_quartet_3.svg)

[3] T. G. Beckwith, R.D. Marangoni, and J. H. Lienhard V, Mechanical Measurements, 6th ed. (Prentice Hall, Upper Saddle River, NJ, 2007), Appendix F (available in the 2.671 Lab).

[4] Frost, J. "Regression Analysis: How Do I Interpret R-squared and Assess the Goodness-of-Fit?", 30 May 2013, http://blog.minitab.com/blog/adventures-in-statistics-2/regression-analysis-how-do-i-interpret-r-squared-and-assess-the-goodness-of-fit, accessed Jan 20, 2018.

[5] T. G. Beckwith, R.D. Marangoni, and J. H. Lienhard V, Mechanical Measurements, 6th ed. (Prentice Hall, Upper Saddle River, NJ, 2007), Chapter 3 (available in the 2.671 Lab).